

Comparison Between Traditional Methods and Proposed VoxShield Architecture

Aspect	Traditional Methods	VoxShield
Input Representation	Handcrafted features	Raw waveform + spectrogram
Feature Extraction	RPS, MGD, MFCC, LFCC, CQCC, F0	wav2vec2 SSL + LFCC/Log-Mel
Learning Strategy	Manual engineering	Automatic learning
Classifier	GMM, SVM, SVDD	Dense Neural Network
Temporal Modeling	Limited/LSTM/GRU	Transformer self-attention
Feature Domains	Single-domain	Dual-domain
Feature Fusion	Rare	SSL + CNN fusion
Attention	Not used	Attention pooling
Detection	Known attacks	Diverse artifacts
Context	Local	Local + Long-range
Adaptability	Redesign needed	Learns from data
Confidence	Real/Fake	Risk bands
Deployment	Research-oriented	Calibrated
Computation	Lower	Higher
Generalization	Limited	Improved robustness

Major Advantages of VoxShield

- 1 No dependence on handcrafted features using wav2vec2 and CNN.
- 2 Uses two complementary audio views: waveform and spectrogram.
- 3 Transformer captures long-range dependencies.
- 4 Attention pooling focuses on suspicious frames.
- 5 Provides probability-based risk outputs.

Simple Summary for Report/Viva

Traditional spoofing detection methods relied on handcrafted acoustic features and classifiers such as GMM and SVM. VoxShield combines wav2vec2 embeddings and spectro-temporal CNN features with Transformer attention and calibrated risk scores for robust and practical deepfake speech detection.